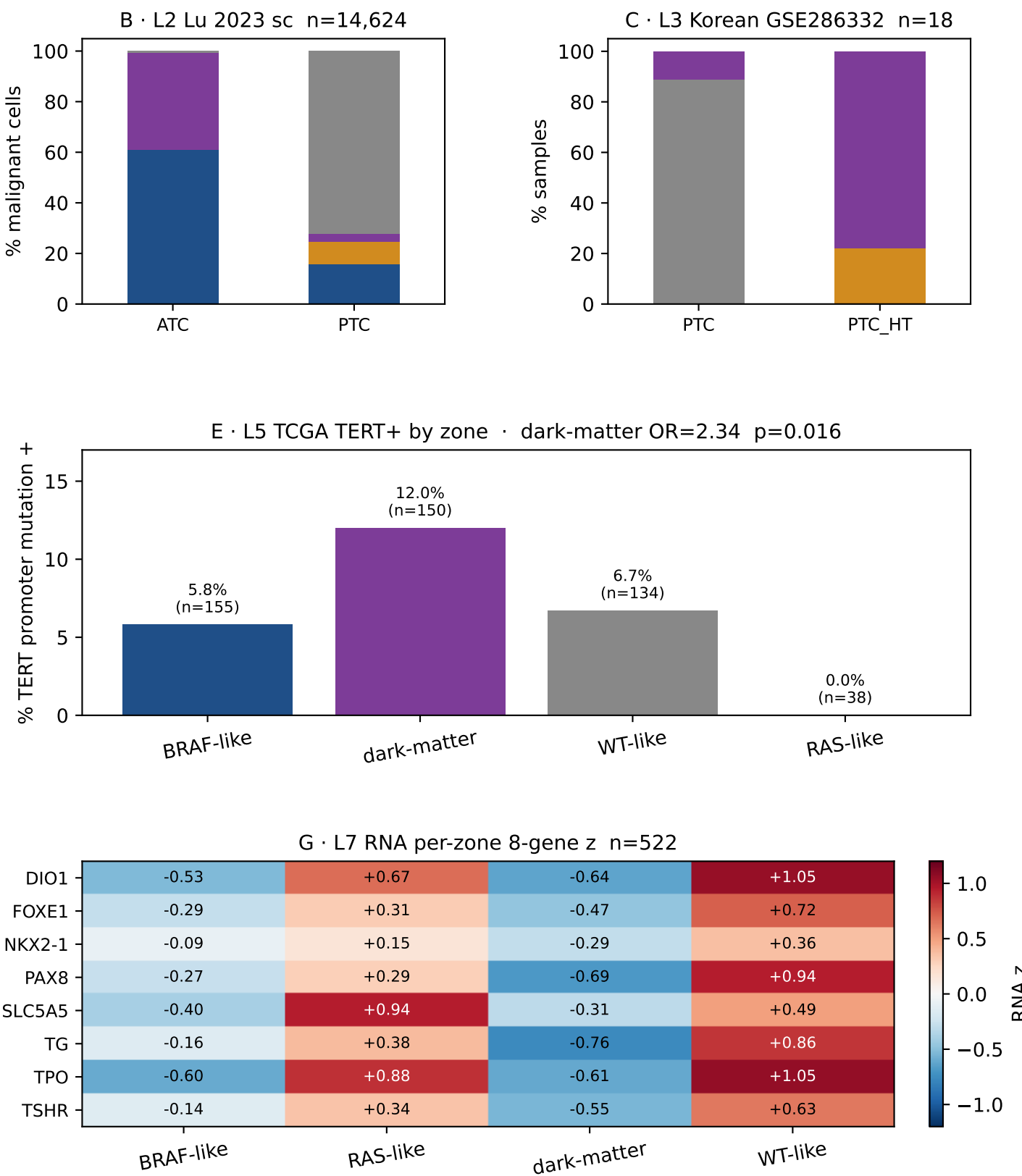
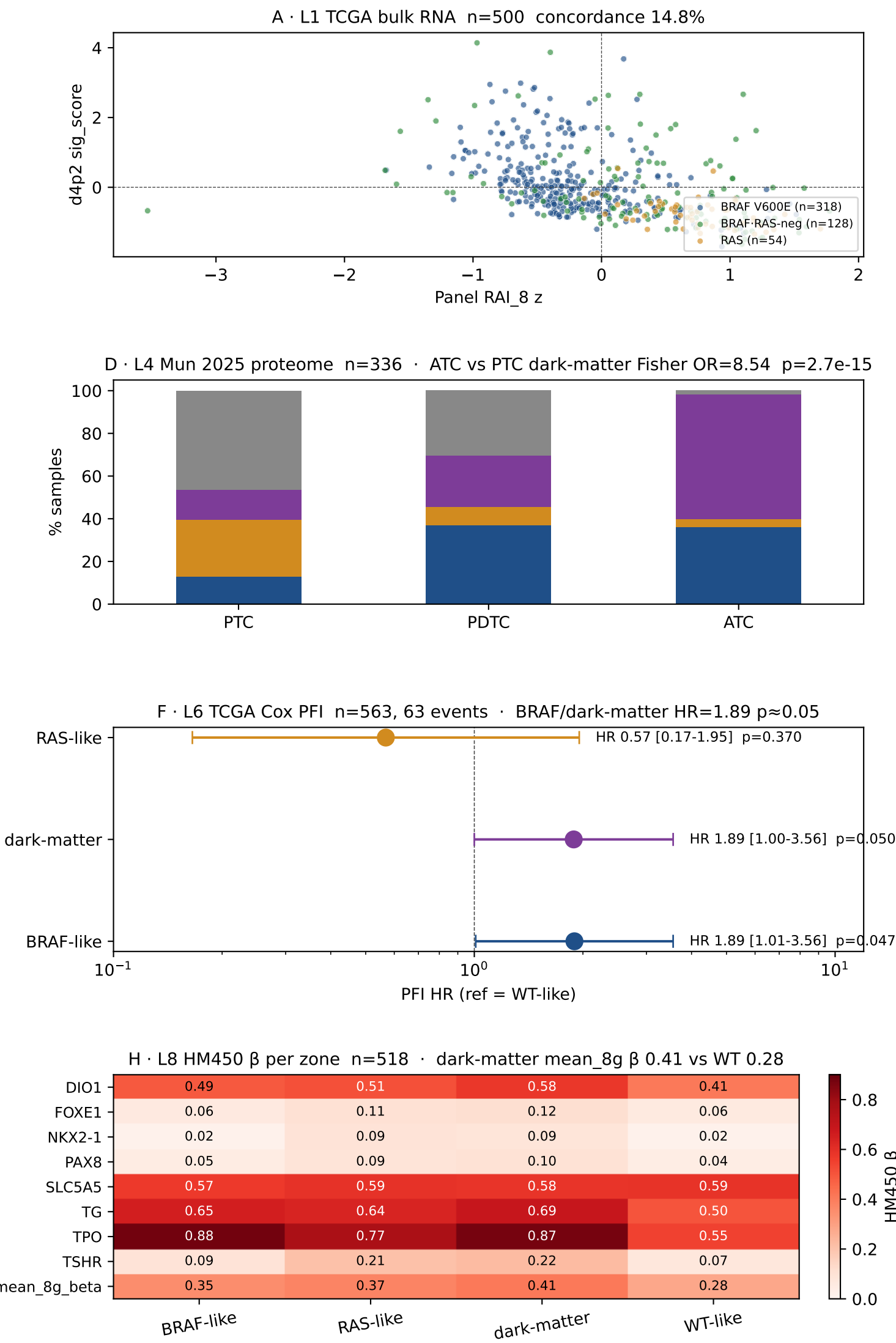




R17 synthesis · 8 layers · 4 pillars (RNA · sc · protein · methylation)

- Two-axis decomposition: HT-overlap (d4p2 sig.DM) vs RAI-lineage (8-gene panel) separates BRAF-like (silenced RAI, no HT), RAS-like (preserved + HT), dark-matter (silenced + HT), WT-like (neither) zones.
- 14.8% RNA concordance is driver-pattern biology, not a labeling bug.
- Dark-matter zone is uniquely:
 - the cellular substrate at sc level (38.3% of ATC malignant cells)
 - the dominant ATC zone at protein level (58.4%, OR=8.54 p=2.7e-15)
 - enriched for TERT+ (OR=2.34 p=0.016) → cellular context for HR=7.57
 - the most hypermethylated (mean 8g β 0.41 vs WT 0.28)
 - the deepest RNA silencing (mean z = -0.54)
 - borderline-prognostic (PFI HR=1.89 p=0.050)

BRAF-like vs dark-matter use different gene-level silencing programs (TPO/DIO1 vs TG/PAX8/TSHR) – methylation-mediated for half the panel.